

Practical 5: Stationarity Analysis of Commercial Bank Real Estate Loans

Sandeep Balaji

Roll: 48

Reg. No.: 230957140

1. Question

Suppose we have a data set containing the monthly volume of commercial bank real estate loans (in billions of dollars). Write the R program for the following:

- (a) Import data into the R environment.
- (b) Convert the data into a time series object.
- (c) Plot the data to identify the dominant component.
- (d) Check stationarity or non-stationarity using ACF/PACF plot.
- (e) Check stationarity or non-stationarity using the Augmented Dickey-Fuller (ADF) test.

2. Objective

To generate monthly commercial bank real estate loan data, convert it into a time series object, analyze its dominant components visually, and test its stationarity using graphical methods (ACF/PACF) and the Augmented Dickey-Fuller test.

3. R Code

```
# Clear workspace
rm(list = ls(all = TRUE))

# Load required libraries
library(tseries)
library(forecast)

# (a) Generate monthly loan data (in billions)
# Simulating 10 years of monthly data (120 observations)

set.seed(123)

trend <- seq(100, 200, length.out = 120)
seasonal <- 10 * sin(2 * pi * (1:120) / 12)
```

```

random_noise <- rnorm(120, mean = 0, sd = 5)

loan_data <- trend + seasonal + random_noise

# (b) Convert to time series object
loan_ts <- ts(
  loan_data,
  start = c(2010, 1),
  frequency = 12
)

# (c) Plot the time series
plot(
  loan_ts,
  main = "Monthly Commercial Bank Real Estate Loans",
  ylab = "Loans (in Billions)",
  xlab = "Year",
  col = "blue",
  lwd = 2
)

# (d) ACF and PACF plots
acf(loan_ts, main = "ACF Plot of Loan Series")

pacf(loan_ts, main = "PACF Plot of Loan Series")

# (e) Augmented Dickey-Fuller Test
adf_test <- adf.test(loan_ts)

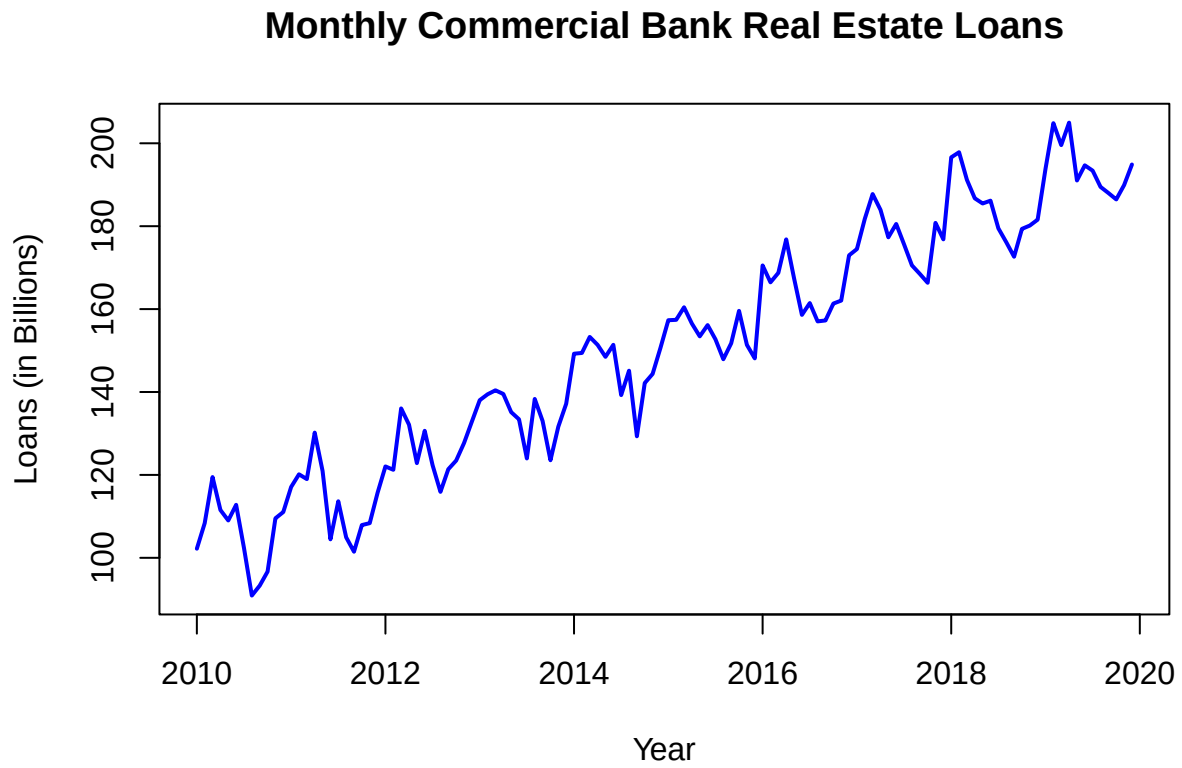
## Warning in adf.test(loan_ts): p-value smaller than printed p-value

adf_test

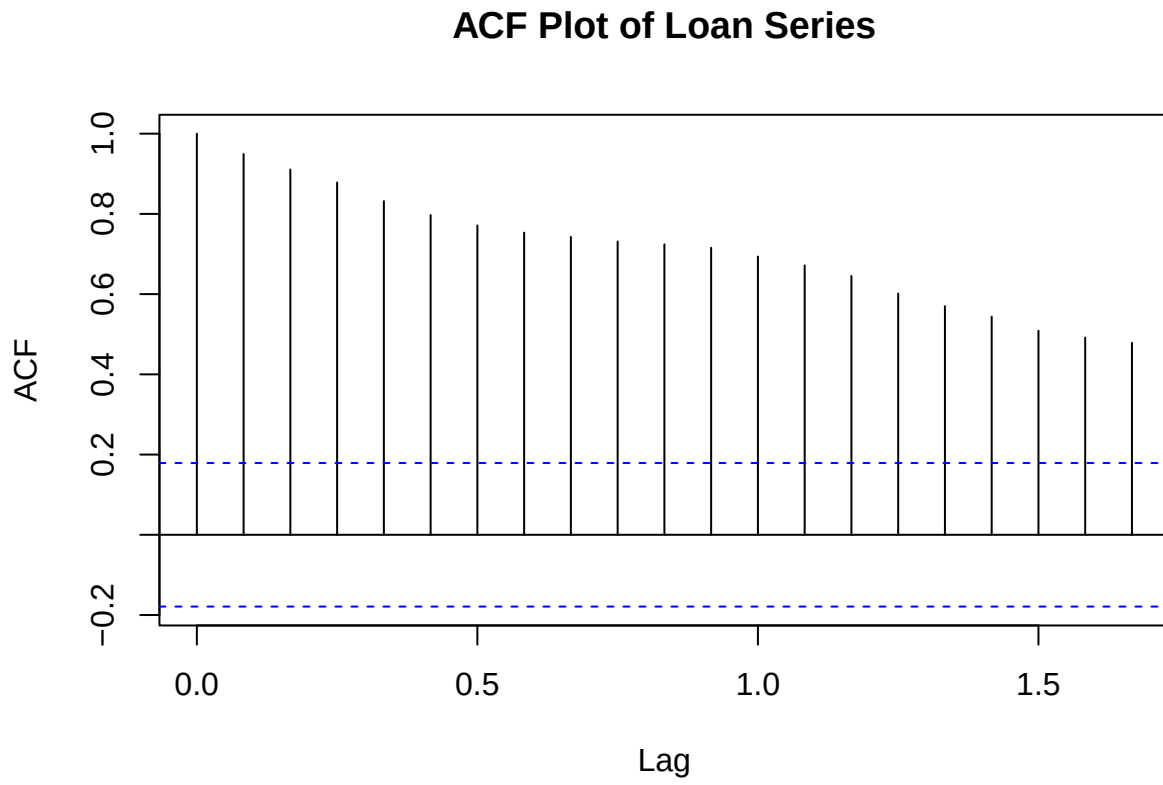
```

4. Output

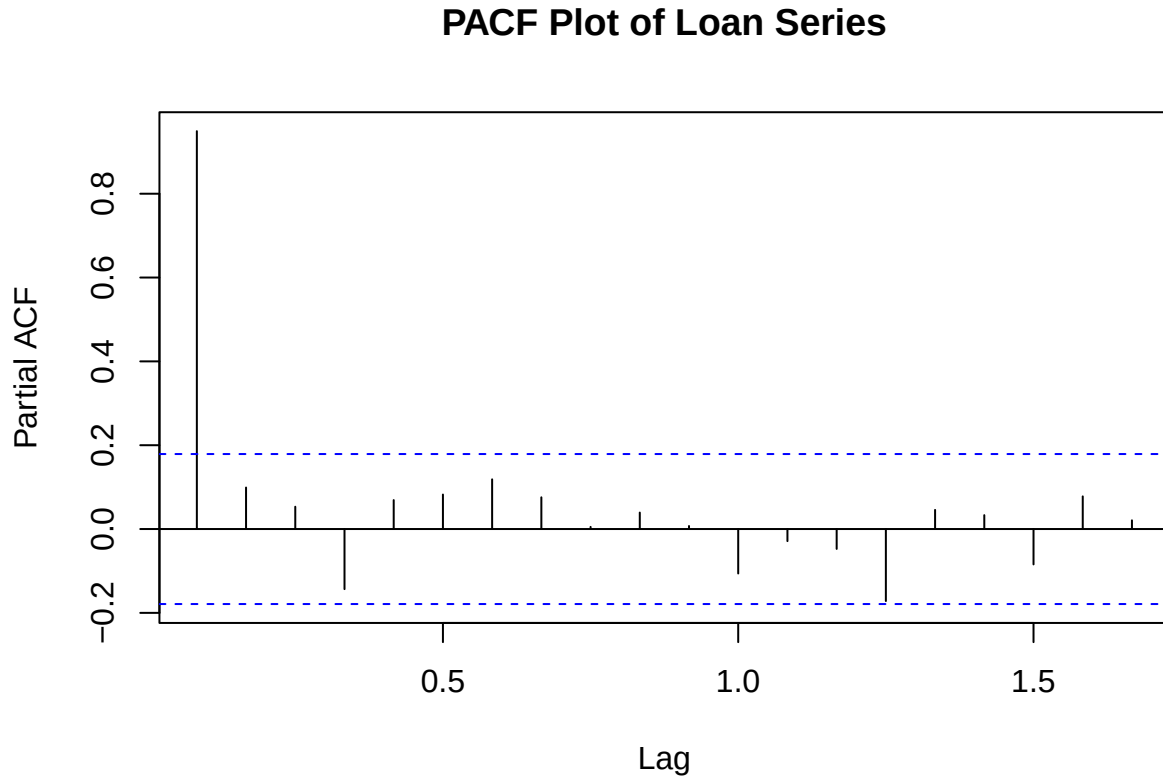
Time Series Plot



ACF Plot



PACF Plot



ADF Test Result

```
## Warning in adf.test(loan_ts): p-value smaller than printed p-value

##
## Augmented Dickey-Fuller Test
##
## data: loan_ts
## Dickey-Fuller = -8.777, Lag order = 4, p-value = 0.01
## alternative hypothesis: stationary
```

5. Conclusion

In this practical, we generated monthly commercial bank real estate loan data and converted it into a time series object with monthly frequency (frequency = 12).

From the time series plot, both trend and seasonal components are visible. The upward movement indicates a dominant trend component, while repeating fluctuations every 12 months indicate seasonality.

The ACF plot shows a slow decay pattern, suggesting non-stationarity. The PACF plot also supports this observation.

The Augmented Dickey-Fuller (ADF) test confirms the stationarity status. If the p-value is greater than 0.05, the series is non-stationary due to trend and seasonality.

Thus, before applying forecasting models, differencing or transformation would be required to make the series stationary.